



Stock Market Analytics Dashboard

Technical Indicators & Financial Terms Guide

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1. Introduction

Technical Analysis is a method used to evaluate stocks by analyzing historical price and volume data. Instead of studying company fundamentals, technical analysts focus on price movements, trading volume, and statistical indicators to identify market trends and trading opportunities.

This document explains the financial concepts and technical indicators used in the Stock Market Analytics Dashboard.

2. OHLCV Data

Open Price

The first traded price of a stock during a trading session.

Example: Market opens at ₹2500. Open Price = ₹2500

High Price

The highest price reached during a trading period.

Example: Price moved from ₹2500 to ₹2528. High = ₹2528

Low Price

The lowest price during a trading session.

Example: Price dropped to ₹2492. Low = ₹2492

Close Price

The final traded price before the market closes.

Example: Market closes at ₹2518. Close Price = ₹2518

Volume

Volume represents the total number of shares traded during a period.

High volume indicates strong market participation.

Low volume indicates weak participation.

3. Moving Averages

Moving Averages smooth price fluctuations and help identify trends.

MA20

Average closing price of the last 20 periods.

Purpose:

- Detect short-term trends
- Dynamic support and resistance

MA50

Average closing price of the last 50 periods.

Purpose:

- Medium-term trend analysis
- Popular among swing traders

MA200

Average closing price of the last 200 periods.

Purpose:

- Long-term trend identification
- Widely used by institutional investors

Golden Cross

A Golden Cross occurs when:

MA50 crosses ABOVE MA200

Meaning:

- Strong bullish signal
- Possible beginning of an uptrend

Death Cross

A Death Cross occurs when:

MA50 crosses BELOW MA200

Meaning:

- Strong bearish signal
- Possible beginning of a downtrend

4. Momentum Indicators

Momentum indicators measure the speed and strength of price movements.

RSI (Relative Strength Index)

Range: 0-100

Interpretation:

- Above 70 → Overbought
- Below 30 → Oversold

Purpose: Measures buying and selling momentum.

MACD (Moving Average Convergence Divergence)

MACD consists of:

- MACD Line
- Signal Line
- Histogram

Bullish Signal: MACD crosses above Signal Line.

Bearish Signal: MACD crosses below Signal Line.



MACD Histogram: Shows the difference between the MACD Line and Signal Line. Positive Histogram: Bullish momentum. Negative Histogram: Bearish momentum.

5. Oscillators

Oscillators identify overbought and oversold market conditions.

Stochastic Oscillator

Contains: %K Line, %D Line

Range: 0-100

Interpretation: Above 80 →
Overbought | Below 20 →
Oversold

Williams %R

Range: 0 to -100

Interpretation: Above -20 →
Overbought | Below -80 →
Oversold

Money Flow Index (MFI)

MFI combines: Price and Volume

Range: 0-100

Interpretation: Above 80 →
Overbought | Below 20 →
Oversold

Often called the **Volume-Weighted RSI**.

6. Volatility Indicators

Volatility measures how much prices move. High volatility means larger price swings. Lower volatility means stable prices.



ATR (Average True Range)

ATR measures average market volatility.

High ATR: Large price movement.

Low ATR: Small price movement.

ATR does NOT indicate trend direction.



Daily Trading Range

Formula: $\text{Daily Range} = \text{High} - \text{Low}$

A larger range indicates greater market volatility.



Rolling Volatility

Calculated using historical returns.

Common periods: 5, 10, 20

Higher values indicate greater market uncertainty.

7. Candlestick Analysis

Bullish Candle

Condition: $\text{Close} > \text{Open}$

Meaning: Buyers controlled the market.

Bearish Candle

Condition: $\text{Close} < \text{Open}$

Meaning: Sellers controlled the market.

Doji Candle

Condition: $\text{Open} \approx \text{Close}$

Meaning: Market indecision.

Candle Body

Formula: $\text{Body} = |\text{Close} - \text{Open}|$

Large Body: Strong buying or selling pressure.

Small Body: Weak momentum.

Upper Wick

Formula: $\text{High} - \text{Max}(\text{Open}, \text{Close})$

Represents selling pressure.

Lower Wick

Formula: $\text{Min}(\text{Open}, \text{Close}) -$

Low Represents buying pressure.

8. Market Statistics

BullishCandle%

Percentage of bullish candles. Higher percentage indicates positive market sentiment.

BearishCandle %

Percentage of bearish candles. Higher percentage indicates negative market sentiment.

Average Candle Body

Measures the average strength of price movement. Large values indicate stronger market momentum.

Average Upper Wick

Shows average rejection from higher prices.

Average Lower Wick

Shows average buying support from lower prices.

Average Return by Trading Day

Analyzes which weekday provides better average returns.

Example: Monday → 0.12% |
Tuesday → -0.08%

Average Return by Trading Hour

Shows which trading hours produce stronger returns. Useful for intraday trading strategies.

9. Trading Terms

Trend

Overall market direction. Types: Bullish, Bearish, Sideways

Resistance

A price level where selling pressure prevents prices from rising further.

Momentum

Measures the strength and speed of price movement.

Risk

The possibility of losing money in a trade.

Support

A price level where buying pressure prevents prices from falling further.

Volatility

Measures how rapidly prices change. High volatility means higher risk and higher reward.

Liquidity

Ease of buying or selling an asset without affecting its price. Higher liquidity generally means lower transaction costs.

Reward

The potential profit from a trade.